higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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This marking guideline consists of 11 pages.

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QUESTION 1

1.1
$$y=\lim_{x\to\infty}\frac{3x^2+e^x}{4x^3+5x} \qquad \frac{\infty}{\infty}$$
$$=\lim_{x\to\infty}\frac{6x+e^x}{12x^2+5} \qquad \frac{\infty}{\infty}$$
$$=\lim_{x\to\infty}\frac{6+e^x}{24x} \qquad \frac{\infty}{\infty}$$
$$=\lim_{x\to\infty}\frac{e^x}{24}$$
$$=\infty$$ (4)

1.2
$$f(x)=sec\,x=\frac{1}{cosx}$$
$$cosx=0$$
$$x=90° \quad \text{or} \quad x=270°$$ (2)
[6]

QUESTION 2

2.1 2.1.1
$$y=\ln^3\left[x^2-5x+\sqrt{x^2-7x+20}\right]$$
$$\frac{dy}{dx}=3\ln^2\left[x^2-5x+\sqrt{x^2-7x+20}\right]\times\frac{1}{x^2-5x+\sqrt{x^2-7x+20}}\times\left[2x-5+\frac{1}{2}\left(x^2-7x\right.\right.$$ (5)

2.1.2
$$y=sec^{-1}\left(x^2\right)+\tan^{-1}\left(\frac{1-x}{1+x}\right)$$
$$\frac{dy}{dx}=\frac{1}{\sqrt{x^4-1}}\times 2x\times\frac{1}{1+\left(\frac{1-x}{1+x}\right)^2}\times\frac{(1+x)(-1)-(1-x)}{(1+x)^2}$$ (5)

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2.2 $y = 3^{x \ln x^2}$

$$\ln y = x \ln x^2 \ln 3$$

$$\frac{1}{y}\frac{dy}{dx} = \ln x^2 \ln 3 + 2 \ln 3$$

$$\frac{dy}{dx} = y\left[\ln x^2 \ln 3 + 2 \ln 3\right]$$

$$= 3^{x \ln x^2}\left[\ln x^2 \ln 3 + 2 \ln 3\right] \quad (4)$$

2.3 2.3.1

$$x^2 y + \cos\left(\frac{\pi}{2} y\right) - 4x^2 = 5$$

$$2xy + x^2\frac{dy}{dx} - \sin\left(\frac{\pi}{2} y\right) \times \frac{\pi}{2} \times \frac{dy}{dx} - 8x = 0$$

$$\frac{dy}{dx}\left(x^2 - \frac{\pi}{2}\sin\left(\frac{\pi}{2} y\right)\right) = 8x - 2xy$$

$$\frac{dy}{dx} = -\frac{8x - 2xy}{x^2 - \frac{\pi}{2}\sin\left(\frac{\pi}{2} y\right)} \quad (3)$$

2.3.2

$$\frac{dy}{dx} = \frac{8 \times 1 - 2 \times 1 \times 2}{2^2 - \frac{\pi}{2}\sin\left(\frac{\pi}{2} \times 2\right)} = 1$$

$$y = 1x + c$$

$$2 = 1(1) + c$$

$$c = 1$$

$$y = x + 1 \quad (3)$$

[20]

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QUESTION 3

3.1 3.1.1 $f(x)=x^3-3x+5$

$f'(x)=3x^2-3=0$

$x=-1 \quad \vee \quad x=1$

$y=7 \qquad y=3$

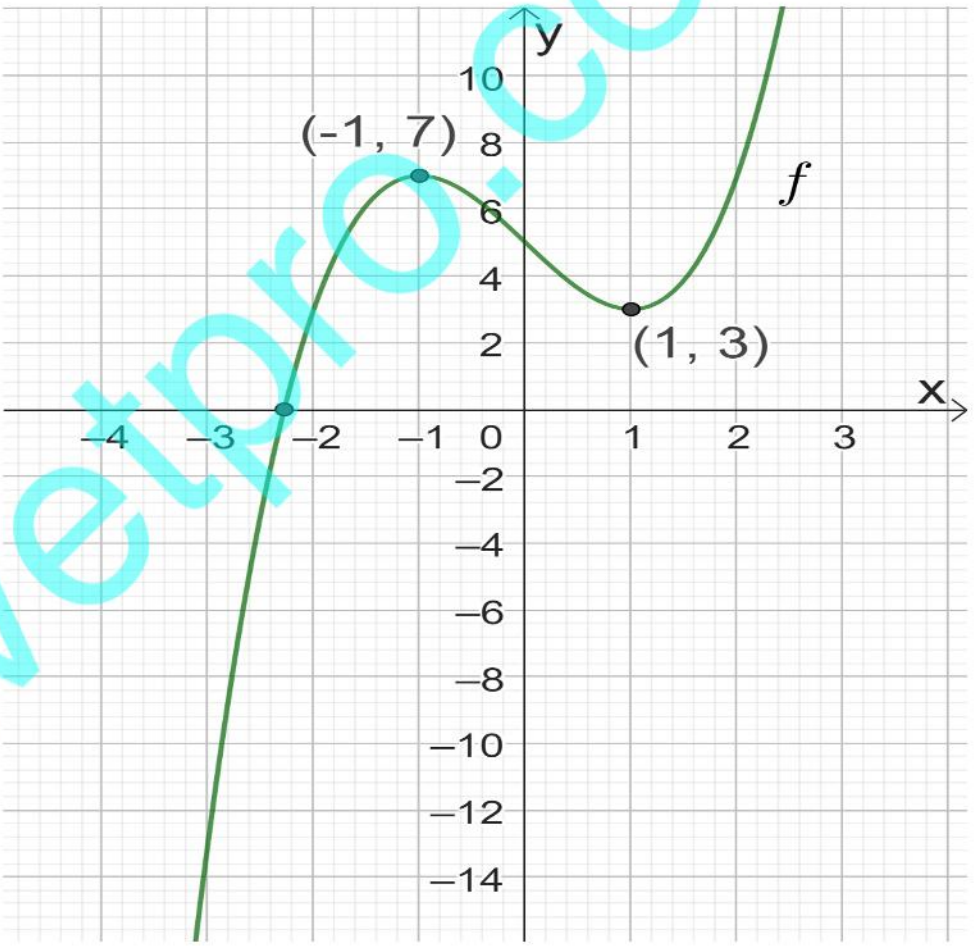
Thus, the coordinates of the turning points are $(-1;7)$ and $(1;3)$. (3)

3.1.2

x	-3	-2	-1	0	1	2
y	-13	3	7	5	3	7

= 1 mark for every 3 correct answers (2)

3.1.3



=1 mark for the shape
=1 mark for the indication of the turning points on the graph (2)

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3.1.4 *Let* $x_0 = -2,3$

$f(-2,3) = -0,267$

$f'(-2,3) = 12,87$

$x_1 = -2,3 + \frac{0,267}{12,87}$

$= -2,279$

$f(-2,279) = -0,003$

$f'(-2,279) = 12,585$

$x_2 = -2,279 + \frac{0,003}{12,585}$

$= -2,279$ (4)

3.2 3.2.1

$s = \frac{1}{3}t^3 - 2t^2 + 3t$

$s = \frac{1}{3}(2)^3 - 2(2)^2 + 3(2)$

$= \frac{2}{3}\ m$ (2)

3.2.2

$s = \frac{1}{3}t^3 - 2t^2 + 3t$

$s' = t^2 - 4t + 3$

$s'(4) = 3\ m/s$ (2)

3.2.3

$s = \frac{1}{3}t^3 - 2t^2 + 3t = 0$

$t(t-3)^2 = 0$

$t = 0\ s \wedge t = 3\ s$ (3)

3.2.4

$s'' = 2t - 4$

$s''(4) = 4\ m/s^2$ (2)

[20]

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QUESTION 4

4.1 4.1.1

$$\int \frac{x(2-3x)}{e^{3x}+x^2}\,dx$$

$$let\ \ u=1+x^2e^{-3x}$$
$$\frac{du}{dx}=xe^{-3x}(2-3x)$$
$$dx=\frac{1}{xe^{-3x}(2-3x)}\,du$$

$$=\int \frac{x(2-3x)}{e^{3x}+x^2}\times\frac{1}{xe^{-3x}(2-3x)}\,du$$
$$=\int \frac{1}{1+x^2e^{-3x}}\,du$$
$$=\int \frac{1}{u}\,du$$
$$=\ln u+C$$
$$=\ln\left(1+x^2e^{-3x}\right)+C \qquad (3)$$

4.1.2
$$y=\cos^7 x\sin^3 x$$
$$=\int \cos^7 x\left(1-\cos^2 x\right)\sin x\ \ dx$$
$$=\int \left(\cos^7 x-\cos^9 x\right)\ \sin x\,dx$$
$$let\ \ u=\cos x$$
$$-du=\sin x\,dx$$
$$=\int \left(u^9-u^7\right)du$$
$$=\frac{u^{10}}{10}-\frac{u^8}{8}+C$$
$$=\frac{\cos^{10}x}{10}-\frac{\cos^8 x}{8}+C$$

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(4)

4.1.3 $\int \cos\left(\frac{\pi}{2}x\right)\sin\left(\frac{\pi}{4}x\right)dx$

$= \int \frac{1}{2}\left(\sin\frac{3\pi}{4}x - \sin\frac{\pi}{4}x\right)dx$

$= -\frac{2}{3\pi}\cos\frac{3\pi}{4}x + \frac{2}{\pi}\cos\frac{\pi}{4}x + c$ (3)

4.1.4 $\int \frac{2x^3 - 5x^2 + 6x + 12}{x-4}dx$

$= \int \left(2x^2 + 3x + 18 + \frac{84}{x-4}\right)dx$

$= \frac{2}{3}x^3 + \frac{3}{2}x^2 + 18x + 84\ln(x-4) + C$ (5)

4.1.5 $\int 4x\cos 4x\,dx$

---

let $u = 4x$ $\qquad dv = \cos 4x\,dx$

$du = 4\,dx$ $\qquad v = \frac{1}{4}\sin 4x$

$\int x\cos 4x\,dx = 4x\sin 4x - \int \sin 4x\,dx$

$= x\sin 4x + \frac{1}{4}\cos 4x + C$ (4)

4.2 $\int \frac{x-5}{x^2+5x+4}$

$\frac{x-5}{(x+1)(x+4)} = \frac{A}{x+1} + \frac{B}{x+4}$

$x - 5 = A(x+4) + B(x+1)$

---

let $x = -4$; $\qquad B = 3$

$x = -1$; $\qquad A = -2$

$= \int \left(\frac{-2}{x+1} + \frac{3}{x+4}\right)dx$ (5)

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$$= -2\ln(x+1)+3\ln(x+4)+C$$

[24]

QUESTION 5

5.1
$$= \int_0^{\infty} \frac{7}{3} t\, e^{-st}\, dt$$

$$let \;\; u = t \;\; \wedge \;\; dv = e^{-st}\, dt$$

$$du = 1\, dt \;\; \wedge \;\; v = -\frac{1}{s} e^{-st}$$

$$= \frac{7}{3}\left(-\frac{t}{s} e^{-st} + \frac{1}{s}\int e^{-st}\, dt\right)$$

$$= \frac{7}{3} \lim_{a \to \infty} \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st}\right]_o^a$$

$$= \frac{7}{3}\left((0-0)+\frac{1}{s^2}\right)$$

$$= \frac{7}{3s^2}$$ (5)

5.2 5.2.1
$$1 - x = \sqrt{1-x^2}$$

$$1 - 2x + x^2 = 1 - x^2$$

$$2x^2 - 2x = 0$$

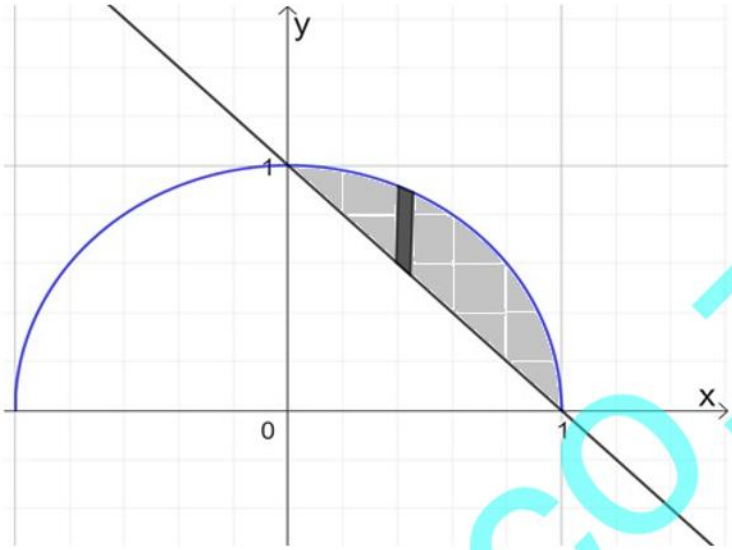
$$2x(x-1) = 0$$

$$x = 0 \quad \vee \quad x = 1$$
$$y = 1 \qquad\qquad y = 0$$

Thus, the coordinates of the points of intersection for the bounded area are $(0;1)$ and $(1;0)$. (3)

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5.2.2



= 1 mark for indication of each enclosed area
= 1 mark for indication of the vertical or horizontal strip (2)

5.2.3

$$A=\int_a^b (y_1-y_2)\,dx$$

$$=\int_0^1 \left[\sqrt{1-x^2}-(1-x)\right]dx$$

$$=\left[\frac{1}{2}\sin^{-1}x+\frac{x}{2}\sqrt{1-x^2}-x+\frac{1}{2}x^2\right]_0^1$$

$$=0{,}285 \ \ units^2$$ (3)

5.2.4

$$V=\pi\int_a^b (y_1^{\,2}-y_2^{\,2})\,dx$$

$$=\pi\int_0^1 \left[\left(\sqrt{1-x^2}\right)^2-(1-x)^2\right]dx$$

$$=\pi\int_0^1 (2x-2x^2)\,dx$$

(4)

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$$= \pi \left[x^2 - \frac{2}{3} x^3 \right]_0^1$$

$$= \frac{\pi}{3} \quad \vee \quad 1{,}047 \ \ units^3$$

5.3 $$m = \ \sigma = \frac{m}{A} = \frac{m}{\pi r^2}$$

$$dm = \sigma dA = \frac{m}{\pi r^2} dA$$

$$d I_z = (\ distance\ )^2 dm$$

$$= x^2 \frac{m}{\pi r^2} dA$$

$$= x^2 \times \frac{m}{\pi r^2} \times 2\pi x \ \ dx$$

$$I = \frac{2m}{r^2} \int_0^r x^3 \, dx$$

$$= \frac{2m}{r^2} \left[\frac{x^4}{4} \right]_0^r$$

$$= \frac{1}{2} m r^2 \ \ unit\, s^4$$

(5)
[22]

QUESTION 6

6.1 $$\frac{dy}{dx} = \frac{3}{y \cos^2 x}$$

$$y dy = 3 \sec^2 x \, dx$$

$$\frac{1}{2} y^2 = 3 \tan x + C$$

$$\frac{1}{2} (2)^2 = 3 \tan 45° + C$$

$$C = -1$$

$$\therefore \frac{1}{2} y^2 = 3 \tan x - 1$$

(5)

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6.2 $$\cos x \frac{d^2y}{dx^2} = \sin 2x + 2^{3x} \cdot \cos x$$

$$\frac{d^2y}{dx^2} = 2\sin x + 2^{3x}$$

$$\frac{dy}{dx} = -2\cos x + \frac{2^{3x}}{3\ln 2} + A$$

$$y = -2\sin x + \frac{2^{3x}}{9\ln^2 2} + Ax + B$$

(3)
[8]

TOTAL: 100

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